PROJECT SYNOPSIS

FAKE REVIEW DETECTION SYSTEM

A Machine Learning and Natural Language Processing based project for classifying online product reviews.

1. Project and Team Details

Student Name	College Name	Branch	Mobile No.	Mail ID	Role
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Domain: Machine Learning (with Natural Language Processing).

Guide Details: Jyoti Prajapati Mam (Project Guide).

Working Project Title

Fake Review Detection System Using TF-IDF and Logistic Regression

2. Problem Statement

Online product reviews can influence purchasing decisions, while some reviews may be misleading, artificially generated, or otherwise unreliable. Manually checking a large number of reviews is time-consuming. This project therefore aims to build an automated text-classification system that provides a preliminary prediction of the class associated with a review. The supplied dataset contains 40,432 reviews across 10 product categories, with ratings from 1 to 5 and two target classes, CG and OR. The project focuses on cleaning review text, extracting useful textual features and applying supervised machine learning for classification.

3. Objectives

• Develop a machine learning system for classifying reviews into the supplied CG and OR classes.

• Clean and normalize review text before model training.

• Convert review text into numerical features using TF-IDF.

• Train a Logistic Regression classifier for binary text classification.

• Evaluate the classifier using a held-out test set.

• Provide a foundation for an interactive review-classification interface using Streamlit.

4. Methodology / Implementation Plan

Overall workflow: Review Input → Text Preprocessing → TF-IDF Vectorization → Logistic Regression → Prediction → Output.

• Dataset inspection and validation.

• Text preprocessing, including string conversion, lowercasing during TF-IDF processing, tokenization and English stop-word removal.

• TF-IDF feature extraction with a maximum of 10,000 features.

• Train Logistic Regression with max_iter=1000.

• Use an 80/20 stratified train-test split for reproducible evaluation.

• Evaluate the model using accuracy, precision, recall, F1-score and a confusion matrix.

• Optional deployment through a Streamlit interface for entering a review and receiving a predicted CG/OR class.

5. Resources and Timeline

A. Hardware & Software Requirements

• Laptop/PC with sufficient memory for text vectorization and model training.

• Python programming language.

• Pandas and NumPy for data handling.

• Scikit-learn for TF-IDF and Logistic Regression.

• Jupiter Notebook / VS Code / Google Colab for development.

• Streamlit for optional interactive deployment.

B. Dataset

Dataset availability: Available. The supplied fake reviews dataset contains 40,432 rows and 4 columns: category, rating, label and text_. There are no missing values in these observed columns. The rating field ranges from 1 to 5, and the two label classes are balanced with 20,216 CG and 20,216 OR reviews. The dataset covers 10 product categories.

C. Project Timeline

Phase	Task	Tentative Completion Date
Phase 1	Dataset collection, inspection and problem understanding	Week 1
Phase 2	Text preprocessing and TF-IDF feature extraction	Week 2
Phase 3	Logistic Regression model training	Week 3
Phase 4	Evaluation, confusion matrix and result analysis	Week 4
Phase 5	Streamlit interface / prototype and documentation	Week 5

6. Expected Outcomes / Conclusion

The expected outcome is a functional baseline machine-learning system capable of classifying online product reviews into the supplied CG and OR classes. The supplied report shows that the reproducible TF-IDF + Logistic Regression approach achieved 86.46% accuracy on an 80/20 held-out test set of 8,087 reviews. The class-wise precision, recall and F1-score were 0.8740, 0.8521 and 0.8629 for CG, and 0.8557, 0.8771 and 0.8663 for OR. The system is intended as a classification aid, not as an independent verifier of a review's truthfulness. Future work can include larger and more diverse datasets, comparison with other classifiers and transformer-based NLP models, confidence scores, category-wise error analysis and Streamlit deployment.

7. References

• Supplied fake reviews dataset (fake reviews dataset.csv).

• Supplied project report: Fake Review Detection System.

• Scikit-learn documentation — TfidfVectorizer.

• Scikit-learn documentation — Logistic Regression.

• Supplied Logistic Regression model (as referenced in the project report).